

**The puzzle of language without syntax:
A review of Steven Mithen’s (2024) “The Language Puzzle”**

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Abstract

In his recent (2024) book “The Language Puzzle,” Steven Mithen argues, among other things, that universal grammar has been altogether debunked, and that the question of the origin of natural-language syntax and compositionality has largely been answered by the work of computational evolutionary linguists Simon Kirby and collaborators, thus: “syntax spontaneously emerges from the generation-to-generation language-learning process itself.” I show that Kirby and collaborators’ work in fact *presupposes* the entirety of syntax in an innate language of thought, and that it subscribes to a notion of universal grammar no weaker than the mainstream one from generative linguistics in the twenty-first century. I point out that Mithen’s negative outlook on generative linguistics is quite pervasive among the general scientific community, even if generative linguistics still dominates many *linguistics* departments. I submit that such widespread misconception about the status of key research programs in linguistics has very pernicious consequences for science. I point to the generalist scientific journals’ failure to perform their central function, and to old rifts among linguists of different traditions as noteworthy sources of this problem. I propose that it is high time, after the linguistics wars, for a linguistics détente.

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Almost everybody is interested in human language: language has the remarkable property of allowing us to express a considerable chunk of our mental life, serving many purposes. Given this intuitively miraculous nature of language, it is no surprise that understanding *what* precisely language is and *how* it works has proved an extremely difficult project in cognitive science. Arguably the only thing harder has been to understand the *evolution* of this striking human capacity (Lewontin, 1998).

Steven Mithen tackles this question head-on in his (2024) book *The Language Puzzle* (TLP). A pioneer of cognitive archaeology, Mithen has written many books on human history and evolution, discussing language in a few of them. For example, in his excellent (1996) book *The Prehistory of the Mind*, Mithen proposes that modern-human thought, likely in tandem with its expression in language, achieved complete integration of three cognitive modules which we find encapsulated in our modern primate cousins: a cognitive module handling social relations, one for intuitive biology, and one for making and using tools. He surveys the cultural and technological capacities of our ancestors as they can be surmised from the archaeological record, and advances specific hypotheses for when *partial* integration of subsets of these modules might have occurred.

But TLP is Mithen’s first deep dive into human language, what it is and how it evolved. My remarks here are concerned exclusively with Mithen’s treatment of syntax, compositional semantics, and universal grammar. In a nutshell, Mithen effectively excludes syntax and compositional semantics from the language puzzle, taking the core of the puzzle to be about *words* and *naming*. Additionally, he summarily discards all linguistic work that openly acknowledges reliance on a notion of universal grammar. I review the literature which Mithen credits with deflating the puzzles of syntax and compositional semantics, and show that it does no such thing. I argue that, rather than eschewing universal grammar, this work presupposes a notion of universal grammar no simpler than the universal grammar characteristic of generative research in the twenty-first century.

Mithen presents a very bleak outlook on generative linguistics which is in no way unique to him. While generative linguistics is perhaps still hegemonic in the wealthiest *linguistics* departments in many countries, the general scientific community largely sees it as an obsolete framework, at best of historical interest, at worst a fundamentally wrong program which has been stymieing progress in

the language sciences since shortly after its inception.

I conclude this piece with speculative but I hope useful considerations about how such unwarranted negative views of generative linguistics have become so prevalent in the general scientific community. I propose that the generalist scientific journals are failing to perform their function, and that old rifts within linguistics have been extremely detrimental to the field and to science. I believe that the first step toward ameliorating these problems is for each of us linguists to listen carefully to what colleagues from other traditions are saying. I am convinced that what unites us is far more than what divides us, and that our seemingly most virulent disagreements are largely spurious.

Syntax and compositional semantics

Mithen dedicates a relatively brief chapter to natural-language syntax and compositional semantics. This is because he takes it that the question of the origin of syntax and compositional semantics has largely been answered by computational evolutionary linguistics, specifically the work of the Edinburgh lab of Simon Kirby and collaborators, which I will review presently (following Mithen, I will use "Edinburgh" as a metonym for this lab). To be clear, Mithen does not say that there are no more live questions about the nature of syntax or the details of its evolution. The view is simply that the results amassed by the Edinburgh group over the past two decades provide an answer that is complete enough for present purposes to the fundamental question of the origin of syntax and compositional semantics, thus: "syntax spontaneously emerges from the generation-to-generation language-learning process itself" (TLP p. 181).

The Edinburgh studies use agent-based models to simulate language learning. In very broad strokes, the models are as follows. Agents begin with no knowledge of an external language, no grammar. Their task is to communicate randomly-sampled meanings to other agents. Each agent first tries to use the grammar it has, and if its grammar offers no way of expressing the target meaning, the agent will innovate by a partly random process. Thus, agents at generation-zero can only innovate, and end up producing holistic utterances with no syntax. Through a mechanism of grammatical subsumption, the generation- $n + 1$ agent, learning from a generation- n speaker, will attempt to collapse two or more rules from the grammar it is getting. For example, an agent could have a rule that generates "Mary snores" and another that generates "John snores." The subsumption mechanism allows an agent to invent a new rule that abstracts away "Mary/John" to

have a single, general rule that generates “ X snores” for a category X containing (at least) “John” and “Mary.” This allows for innovation in rules, and eventually leads to generalization. Now, the exciting result is that, in this setup, a grammar with complex, recursive syntax and compositional semantics will reliably arise and remain quite stable across later generations.

But here is the rub: the Edinburgh simulation studies that show emergence of complex syntax *presuppose* that meanings themselves are linguistically structured.¹ For example, Kirby (2002) gives models where the space of meanings includes such expressions as `says(mary,thinks(mary,praises(john,gavin)))`, meant to have (roughly) the same content as the English sentence “Mary says she thinks that John praises Gavin.” There is simply no other way to interpret this setup: there are *two languages* here, an innately given internal linguistic system for composing complex, hierarchically structured meanings, which one might call a language of thought, and an external linguistic system whose evolution we are simulating.² The external linguistic system will *begin* with entirely holistic utterances, but under certain conditions it will eventually, in fact “inevitably” in Kirby’s own words (2002, pp. 191–192), end up just as compositional and recursive as the language of thought it is meant to express. More concretely, for the language-of-thought expression above, a simulation might settle on a grammar that expresses that meaning as “[that [that gavin praises john] thinks mary] says mary.”³

Since the space of meanings is crucially comprised of expressions in a *language* of thought, it is misleading to characterize these results as showing that syntax in natural language emerges from the language-learning process itself. In actuality, syntax is fully presupposed in the language of

¹ Mithen’s (TLP p. 184 ff.) discussion focuses on the version of this work where meanings are taken to be *strings* of semantic values, as per Brighton et al. (2005) or Ren et al. (2020). But as Brighton et al. (2005, p. 186) explain, this class of models crucially do *not* produce hierarchical syntactic structures. This is exactly as expected: if each meaning is a vector (i.e. a string of values), then language will evolve to mirror that very structure, and will therefore produce strings of symbols with no further hierarchical structure. Mutatis mutandis for other versions of this work that use slightly more complex feature structures like attribute-value pairs. Thus, the very articles Mithen cites acknowledge that a proper explanation of the highly complex syntactic structures routinely found in the languages of the world requires the more complex meaning spaces proposed by Kirby (2000, 2002). Consequently, the Kirby models with more structured meanings are the ones I discuss here.

² Croft (2004) makes observations in the same vein, concluding in particular that “the real question is, where did the ‘language of thought’ come from?” (p. 609).

³ I take some liberty for expository purposes with this example. First, the external-language words in Kirby’s models are arbitrary strings of letters. Second, the strings don’t come with brackets to indicate their constituent structure, but they are the outputs of proper generative grammars, so each string comes with its own parse tree. Third, while some grammars in Kirby’s studies develop what looks like a complementizer (“that” in English), this doesn’t always happen.

thought, and without this assumption syntax will *not* emerge in the external language from the language-learning process. Thus, the Edinburgh results in no way deflate the puzzle of natural-language syntax. Rather, these results push the onus of describing and explaining a key aspect of natural-language syntax and compositional semantics onto an unspecified field of study that is meant to figure out the language of thought.

Now, whose job is it to carry out this crucial research program? Many different kinds of expertise are needed of course, but clearly *linguists*, in particular syntacticians and semanticists, must play a central role in this project. Indeed, the Edinburgh results constitute a simple but insightful validation of Noam Chomsky’s “Cartesian” program for linguistics (Chomsky, 1965, 1966): natural language exists to express thought, in internal calculation (Harman, 1975/1999) as in external communication,⁴ therefore natural language is deeply influenced by the structure of thought. It follows that one especially important way to investigate the structure of thought is to investigate the structure of natural language. Of course, these simulation results, with their external-language syntax that is practically isomorphic to the presupposed thought syntax, are highly idealized. If only our natural languages manifested the structure of thought this straightforwardly! Instead, myriad other factors will contribute to shaping external language, but the basic foundational tenet of generative linguistics remains valid: to investigate language is, in part, to investigate thought.

This is not to say that a theory of thought is the necessary goal of all linguistic investigations. The connection with thought is real and fascinating, but it is not the only interesting thing about language. For example, Everett (2017) is most interested in language as a culturally invented tool, focusing on its communicative function. Everett argues that many languages show no evidence of syntactic recursion, but he happily grants that the *thoughts* of speakers of such languages are fully recursive (p. 31 ff.).

The trouble is that no one knows how to account for the structure of thought (in particular recursion) without treating it as a language (Fodor, 1975, 2008; Quilty-Dunn et al., 2023). The puzzles of language and thought are not independent, as generative linguistics has always held and the Edinburgh simulation studies corroborate. Consequently, while there’s nothing wrong with

⁴ Much if not most scholarly work on these topics is wont to choose one of these two functions and declare it “the primary,” “the real,” or “the original” function of language. I am very puzzled by these singular definite descriptions, and cannot fathom why anyone would think that we cannot move forward without making such a choice.

investigating language without aiming at thought, we cannot summarily discard linguistic investigations that do.

Universal grammar

Mithen makes a few references to universal grammar and dedicates a few pages to both characterizing the view *and* explaining how it has been utterly debunked (TLP pp. 42–45). There are two points I want to make in this connection. First, that Mithen’s arguments against universal grammar do not warrant his summary dismissal of it. Second, that the very work from computational evolutionary linguistics he considers an alternative view in reality presupposes a version of universal grammar that is no weaker (i.e. it does not make fewer commitments) than the mainstream view of universal grammar from generative linguistics in the twenty-first century.

Mithen marshals three orders of arguments against universal grammar. First, he submits that “a language-learning toolkit implies one or more dedicated areas in the brain for language, supported by language-specific genes” (TLP p. 44), and that to date no such language-specific brain areas or genes have been found. A later discussion of language in the brain suggests that what worries Mithen is that “language is distributed throughout the brain rather than being located in a small number of specific areas” (p. 236), and that “no area of the brain appears to be entirely dedicated to language—they are all multifunctional” (p. 237). This is all a little baffling. Mithen seems to demand that psychology be mappable to neurology in ways that would make immediate sense to Paul Broca, but this is not a recognized razor in cognitive science.⁵ And if Mithen’s worry concerns brain areas for language *acquisition* specifically, then it is crucial to remind ourselves that natural language, like so many other psychological capacities in humans and other animals, is characterized by a *critical period* for acquisition. If a child does not encounter the right stimulation from the environment, then there is a point after which it becomes too late to acquire language. It follows that the neurological bases of a language-acquisition device must be investigated in *children’s* brains, not adults’. To my knowledge, developmental neuroscience currently allows for no conclusion either

⁵ A knowledgeable colleague tells me that some generative linguists, specifically “biolinguists,” have staked out positions of this sort in print. This may well be true, but it is irrelevant to my goals here. I am not arguing for the infallibility of scholars who identify as generative linguists, I am arguing that the principles of generative linguistics are nowhere near as silly as they are so often made out to be. Even if some generative linguists have defended naive views of neural organization, this cannot be used to impugn ideas that, in and of themselves, make no particular commitments about how psychological capacities map to neural structures.

way, and Mithen offers no citation suggesting otherwise.

Mithen’s remark on genes fares no better: a requirement that cognitive capacities must be neatly traceable to genes would render both cognitive psychology *and* cognitive neuroscience altogether impossible given our present lack of understanding of the mapping from genes to neural structures, psychological capacities, and complex behavior.

Second, Mithen claims that poverty-of-stimulus arguments have been debunked, referring to advances in our understanding of how children acquire the meanings of words, and to the Edinburgh results on “spontaneous” emergence of syntax. But the acquisition of *words* was never the central target of poverty-of-stimulus arguments from linguistics. The core of the argument was always about syntax and compositional semantics. And these, as I’ve been arguing, are in no way deflated by the Edinburgh results.

Finally, Mithen points to the work of Evans and Levinson (2009) on the scarcity of language universals as a debunking of universal grammar. But *universal grammar* and *language universals* are different notions. Universal grammar, as Mithen correctly states, is the hypothesis that there is non-trivial innate structure dedicated to acquiring and using language. Language universals are generalizations about what is to be found *everywhere* in the languages of the world. Universal grammar predicts the *existence* of language universals, but *which* universals are predicted to exist depends on the particular theory of universal grammar. Evans and Levinson (2009) do not prove or for that matter claim that there are and can be no facts true of all languages. Rather, they argue against a host of *particular* (purported) language universals. Someone would have to check of course, but their conclusions might very well be compatible with versions of universal grammar that are weak enough. Two clear examples of extremely weak theories of universal grammar are the minimalist program which dominates research in generative syntax, more on which shortly, and crucially Kirby’s own universal grammar, as I argue presently.

The models by Kirby (2002) assume that each agent is trying to learn a context-free grammar, thereby significantly limiting the range of learnable languages. Moreover, the agents have a crucial and far-from-trivial mechanism for rule subsumption, as well as other machinery I could not review here. This is a relatively weak notion of universal grammar to be sure, nothing like the principles and parameters framework during its heyday in the 1980s. Nevertheless, an eschewing of

universal grammar, as understood by Mithen and myself, it most certainly is not.

Kirby (2002) characterizes what I am calling his universal grammar as a "language learning device" (LLD), which he contrasts with generative universal grammar's language *acquisition* device (LAD). He submits that the constraints imposed by his LLD "are far less severe than those imposed by the Chomskyan model of the LAD" (2002, p. 196). This contrast was arguably fair for earlier versions of "Chomskyan" universal grammar, but by 2002 it was well on its way to becoming anachronistic. Most generative research since Chomsky's (1995) Minimalist Program has repudiated the highly detailed proposals from the 1980s, despite the protestations of many linguists (see for example Lappin et al., 2000). Hauser et al. (2002) in particular express, as it were, the hope that the binary structure-building operation Merge will suffice in way of a language-specific device. This has been *the mainstream* in "Chomskyan" generative syntax for the past thirty years: try to make do with just Merge, and expect all Merge-independent trends observed in the languages of the world to be attributable to considerations of efficiency of computation and domain-general constraints on cognition. And while perhaps many of these scholars suspect that communicative pressures and learning bottlenecks will make only very peripheral contributions to language, their minimalist program is perfectly compatible with such additions.

I must disclose here that I am at best agnostic as to whether minimalism was a positive development in generative linguistics. Be that as it may, in light of the mainstream generative notion of universal grammar from the twenty-first century, the purported advantage of Kirby's LLD over the generative LAD dissipates entirely.⁶ Indeed, one might have expected that the dramatic weakening of universal grammar proposed by minimalism should at least have made generative linguistics entirely palatable to cognitive scientists interested in as small a language-specific faculty as possible and in a prominent role for domain-general pressures in shaping natural languages. Yet by and large it has had no such effect. I conclude this piece with remarks about this curious state of affairs and

⁶ An important caveat is in order. As every practicing minimalist syntactician knows, you can't actually get very far with just Merge and entirely general constraints of efficiency of computation. Usually, some non-trivial things need to be said about rich lexical features that encode for highly theory-internal properties, like a requirement to be displaced (movement). But then is it disingenuous to say that minimalism is *minimal*? Yes and no. On the one hand, more is indeed needed than Merge to write a minimalist grammar of an actual natural language. But crucially for present purposes, the very same criticism applies to the Edinburgh work: only within highly idealized settings that constitute proofs of concept can we give the impression that our theory is beautifully economical. As usual in science, detailed empirical work is the bane of every simplicity-loving theoretician.

its consequences for the cognitive science of language.

After The Linguistics Wars, time for A Linguistics Détente

I have argued that Mithen illegitimately excludes syntax and compositional semantics from the language puzzle, and that he dismisses universal grammar while endorsing work that presupposes it. But the view that generative linguistics is fundamentally wrong, or at least altogether obsolete, is far from being idiosyncratic to Mithen, in fact it is quite pervasive. If I am right that none of this is apodictic, an interesting question emerges: how did it come to be so popular?

I believe that the answer has an important historical component in the aftermath of the "linguistics wars" (Harris, 1993/2021), waged between generative linguists on the one side and generative semanticists and their descendants, the cognitive linguists, on the other. Like so many acrimonious separations, the two parties got to keep different circles of friends. The generative linguists, especially the interpretive formal semanticists, kept their relationship with a certain kind of high-prestige philosophy. The cognitive linguists got to keep the cognitive psychologists and some of the neuroscientists. Cognitive psychology and neuroscience have dominated cognitive science for a while now. For example, the lion's share of publications in cognitive-science journals comes from these two disciplines (Núñez et al., 2019). Consequently, what one reads in the pages of the top cognitive-science journals or the top generalist journals (Nature, Science, and so on) on the cognitive science of language is jarringly one-sided. Yet, looking to these journals for an update on the state of the art of the cognitive science of language ought to be entirely justified. This is what these journals *are for*, and if they are giving serious scholars like Steven Mithen the wrong impression about the status of key research programs in the cognitive science of language, then the journals are failing to perform their function. They are part of the problem.

But so are we linguists: the linguistics wars involved much unnecessary and downright internecine conflict. For example, Joan Bybee and Paul Hopper, in the introduction to their (2001, p. 1) edited volume on usage-based linguistics, point out that a premise of structural linguistics preserved in generative linguistics is that "the study of structure is a higher calling than the study of usage." I agree with their assessment: many generative linguists have been and still are incomprehensibly dismissive of facts like statistical frequencies of linguistic constructions in use, which usage-based linguists very plausibly take to be highly informative of how some linguistic

constructions are stored in and accessed from memory.

Of course, if competence-minded linguists have no grounds on which to be disdainful of usage, the converse is also true. And the best cognitive linguists make no such mistake. For example, Bill Croft gives a remarkable *competence* theory of event conceptualization in his tour de force (2012) book *Verbs*. Croft describes in detail a system for representing events as changes of state along time, and he does so in order to characterize the space of possible event conceptualizations in dimensions relevant to thought as inferred from studying natural language.⁷ Usage-based considerations enter the stage *after* this fundamental competence theory has been expounded, for example to explain the tension in the fact that particular verbs have particular default aspectual contours that are nevertheless coercible with enough context. Croft’s work demonstrates that taking usage seriously is in no way incompatible with taking competence seriously.

But is competence-minded generative linguistics relevant for cognitive science more broadly and for the study of the evolution of language, as cognitive linguistics is? Interpretive formal semantics provides a good case for this thesis, though I can only adumbrate it here.

For example, Philippe Schlenker and collaborators have studied in great detail the interactions between syntactic structure and various forms of iconicity (see Schlenker, 2022, chapter 12, for a general scientific audience survey). Spoken utterances can be attended by gestures, as in “If I see Fred, I will reward him [blows a kiss].” It turns out that the non-linguistic component interacts with the syntactic and semantic structure of the sentence in complex ways that can be understood with appeal to generative-linguistics theories of the inferential typology of natural-language utterances (Tieu et al., 2019). This work is particularly relevant to Mithen’s goals in TLP, given the proposed central role for iconicity in language evolution, which I could not review here (but see van Mazijk, 2024).

In conclusion, the schism of the late nineteen-hundreds has been extremely detrimental to our field, and it has engendered great confusion and misconception in the general scientific community regarding what the state of the art in linguistics is. Such misconception has especially pernicious consequences for intrinsically interdisciplinary work of the sort that Mithen conducts in

⁷ Croft presents his theory in graphical terms, which might seem incompatible with the mathematical tools that characterize generative formal semantics, but this is not so: Loreau-Unger (2024) has shown that Croft’s theory can be regimented in the language of general topology.

TLP. If we generative and cognitive linguists cannot work toward an integrated view of our overlapping but different interests in competence and usage, symbolic structure and neural instantiation, truth conditions and metaphorical meaning, to mention just a few, then surely investigations into the evolution of language are doomed to remain shrouded in the dense fog of mystery which Lewontin (1998) predicted would forever be their lot.

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